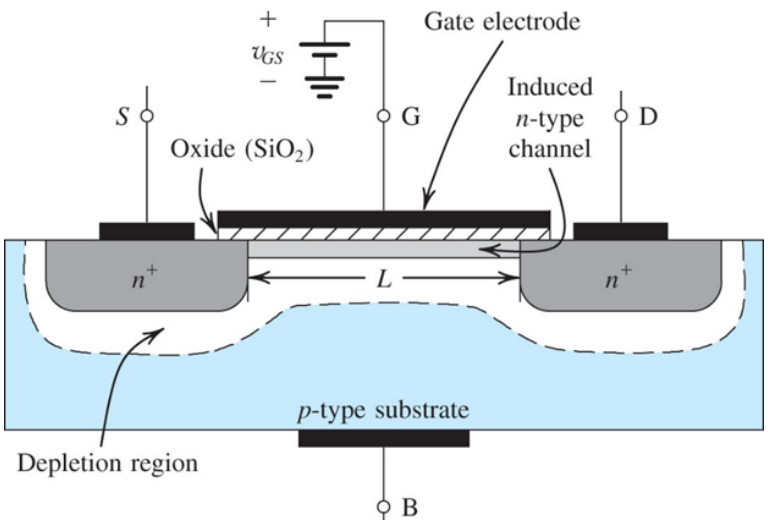
Unit #1
Chapter #5.4: Body Effect and Other Topics
Lecture 4

from **Microelectronic Circuits** Text
by Sedra and Smith
Oxford Publishing

5.4.1 The Body Effect (n-channel MOSFET)

- In some cases we will see, particularly when the NMOS is used as a switch, the body voltage will be less than the source voltage, i.e. $V_{SB} > 0$ V
- This is caused by an increase in fixed charge in the depletion region
- This will lead to a body effect, which cause V_t to increase by

$$\Delta V_t = \gamma(\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f})$$



5.4.1 The Body Effect (n-channel MOSFET)

- The MOSFET parameter γ is called the body effect parameter

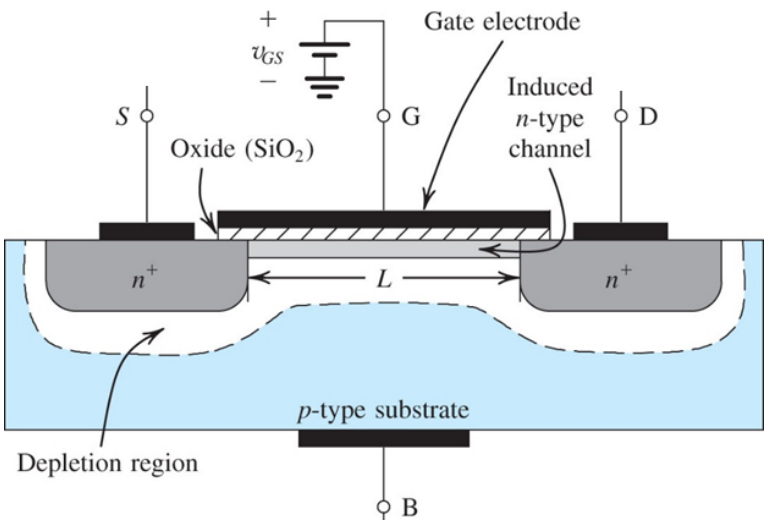
$$\gamma = \frac{\sqrt{2qN_A\varepsilon_s}}{C_{OX}}$$

- The parameter ϕ_f is called the bulk potential.

- With the body effect the threshold voltage becomes

$$V_t = V_{t0} + \gamma(\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f})$$

where V_{t0} is the threshold voltage for V_{SB} = 0V



5.4.2 **Temperature Effects**

- Both V_t and k' are temperature sensitive
- V_t decreases about 2mV for ever 1°C rise in temperature
- k' decrease with temperature and its effect is the dominant one, so overall as temperature is increased the drain current will decrease

Exercise 5.16

An NMOS transistor has V_{t0} = 0.8 V, ϕ_f = 0.35 V, and γ = 0.4 $V^{1/2}$.
Find V_t when V_{SB} = 3V.